Role of Central Banks (CBs)

- CB determine, implement, and control the monetary policy in the country.
- **Fed** is the central bank of the United States of America.
- **4 major functions:** 1) Conduct monetary policy, 2) Supervising & Regulating depository institutions, 3) Maintain stability of fin sys, 4) Source of funds
- **Structure of Fed Reserve:** 12 Fed Reserve Banks, 7-member Board of Gov.
- **Federal Open Market Committee (FOMC)** → major monetary policy making body
  - Req. to meet at least 4 times each year, but meet more often
  - Formulate policies to promote employement, economic growth, price stability
  - Set guidelines regarding open market operations & buy/sell of US govt securities
- **Functions of FRBs:** Assist in conduct of monetary policy, Supervision & Regulation, Consumer protection, Govt. Services, New Currency Issue, Check clearing, Research
- **Fed Liabilities** → Currency in circulation & reserves → Monetary / Money Base
- **Fed Assets** → Treasury & Govt. agency securities, Treasury currency, gold & FX
- **Fed Activities** → Open market operations, Discount Rate, Reserve Requirements
- **Open Market Operations:**
  - Primary determinant of changes in bank excess reserves in the banking system
  - Primarily conducted using Treasury securities, but others can be used as well
- **Discount Rate:**
  - Rate of interest that Federal Reserve Banks charge on loans to FIs in the district
  - Raising discount rates → tightening monetary conditions → higher interest rates
  - Lower discount rates → expansionary monetary conditions → lower interest rates
  - Not used often → Difficult to predict changes in bank discount window borrowing
- **3 Discount Window Lending:** Primary (short term), Secondary (backup), Seasonal
- **Reserve Req.** determine min. amount of reserve assets DIs must maintain.
- **Inc. in RR** → DI must hold more reserve → lend less → less credit avlbl in economy
- **Change in Bank Deposits = Change in Reserves / New Reserve Requirement**
- Fed can successfully target **1 of Money Supply or Interest Rate (Fed Funds Rate).**
- Under current regime, Fed announces if Fed Funds Rate inc, dec, or is same.
- **Fed Funds Rate** is interest rate at which DIs trade excess reserves with each other.
- Independence of Central Bank → Free from political pressure → Long term policies
- **System-wide rescue programs:** Expansion of retail deposit insurance, Capital injections, Debt guarantees, Asset purchases / guarantees

Managing Credit Risk

- Credit Management → Determination of several features of a low-not-dead instrument such as interest rate, maturity, collateral, etc.
- Managerial efficiency and credit risk management directly affect the return and risks of the loan portfolio.
- FIs can diversify some credit risk by exploiting the law of large numbers.
- **Mortgage Loan:**
  - Two considerations dominate F's decision to approve applications
    - Applicant's ability & willingness to make timely interest & principal repayments
    - Value of borrower's collateral
  - **Gross debt service ratio (GDS) = Total Accommodation Expenses / Gross Income** [Acceptable threshold → max 25 - 30%]
  - **Total debt service ratio (TDS) = Total accommodation expenses plus all other debt service payments / Gross Income** [Acceptable → max 35-40%]
  - **Perfecting collateral:** Process of ensuring that collateral used to secure a loan is free and clear to the lender should the borrower default
  - **Foreclosure:** Process of taking possession of the mortgage property if the borrower defaults
  - **Power of Sale:** Process of taking proceedings from the forced sale of mortgage property. And the excess back to the mortgagor.
  - Prior to accepting a mortgage:
    - Confirm title and legal description of property.
    - Obtain surveyor certificate confirming house is within property boundaries.
    - Check with the tax office that no proper taxes are unpaid.
    - Request land title search to determine there are no other claims.
    - Independent appraisal to confirm purchase price is near market value.
- **Consumer and Small Business Lending:**
  - Very similar to mortgage lending. Focus on individual's ability to repay.
  - Small business scoring models combine computer-based financial analysis of financial statements with behavioral analysis of the owner.
- **Mid-market commercial and industrial lending:**
  - Mid-market corporates characterized as:
    - Sales revenue between $5 million and $100 million per year
    - Recognizable Corporate Structure
    - No ready access to deep and liquid capital markets
  - **Short-term commercial loans** are used to finance working capital needs and other short-term funding needs. (maturity < 1 year)
  - **Long-term loans** used to finance credit needs like Purchase of Real Assets, New Venture Start-up Costs, or Inc. in Working Capital (maturity > 1 year)
- **Five Cs of Credit:**
  - **Character** → Probability that the loan applicant will honor loan obligation
  - **Capacity** → Applicant's ability to pay the FI according to loan terms
  - **Collateral** → Assets that loan applicant offers as security backing the loan.
  - **Conditions** → General economic trends that may affect applicants' ability
  - **Capital** → General financial condition of the applicant
- Statement of cash flows separates cash flows into four categories:
  - **Cash flow from operating activities:** Producing and selling products
  - **Cash flow from investing activities:** Buying or selling long-term assets
  - **Cash flow from financing activities:** Debt and equity financing
  - **Net Change in Cash and Marketable Securities** (sum of above three)
- **Ratio Analysis:** Useful when performing financial statement analysis on a mid-market applicant.
  - **Liquidity ratios** express variability of liquid resources.
  - **Asset mngmt ratios** give clues as to how well an applicant uses assets.
  - **Debt and solvency ratios** → asset financing with debt vs. equity.
  - **Profitability ratios** express the profitability of the firm.

Analysis of Performance

- Ultimate measure of CB's performance is the value of its common equity to its shareholders.
- Financial statements are used to analyze performance.
- CAMELS Ratings:
  - **Capital Adequacy:** Volume of risk assets, marginal & inferior assets
  - **Asset Quality:** Distribution & severity of adversely classified assets
  - **Management:** Evaluated against all factors necessary to operate bank safely
  - **Earnings:** Evaluated as per the ability to cover losses and provide adequate capital protection
  - **Liquidity:** Volatility of deposits, frequency and level of borrowings
  - **Sensitivity to Market Risk:** How changes in int. rates, FX, etc. affects earnings / capital
- Balance Sheet presents financial information on bank's assets, liabilities & equity capital
- Income Statement presents categories of revenue and expenses, and net profit over time
- **Assets:**
  - Cash and due from other depository institutions or Fed
  - Investment Securities → Fed funds sold, repos, US Treasury secs
  - Loans and Leases → major assets & largest flow of income, least liquid asset
  - Other assets → trading assets, premises & fixed assets, intangible assets
- **Liabilities:**
  - Deposits
    - Negotiable Order of Withdrawal (NOW) acc. pay interest when min balance is maintained.
    - Money Market Deposit Acc (MMDA) have retail savings account and some limited checking account features
    - Retail certificates of deposits (CDs) are time deposits with a face value below $100,000
    - Wholesale certificates of deposits (CDs) are time deposits with a face value above $100K
  - Borrowed funds → Federal funds, repos, bankers' acceptance, commercial paper, etc.
  - Other liabilities → Accrued interest, deferred taxes, etc.
  - Equity Capital → Preferred and common stock, Retained Earnings, Surplus Capital
- Off-Balance Sheet (OBS) Assets & Liabilities
  - Contingent assets and liabilities that may affect future status of FI's balance sheet.
  - **Loan Commitments** → Contractual commitments to loan a firm certain max amount
  - **Letters of Credit** → Contingent guarantees sold by an FI to underwrite trades
  - **Loans Sold** → Loans originated by the bank and then sold to other investors
    - Recourse is the ability to put an asset or loan back to the seller if credit quality deteriorates.
  - **Derivatives** → Futures forward swap and option positions for hedging or speculation
- **Other fee-generating activity:** Trust Services, Processing (Data) services, Correspondent Banking
- Income statement: Interest Income, Interest Expense, Provision for loan losses, Non-interest income, Non-interest expense, Extraordinary items, Income Taxes
- **Net Interest Income = Interest Income - Interest expense**
- **Income Before Taxes and Extraordinary Items = Net Interest Income - Provision for loan losses + Non-interest income - Non-interest expense**
- **Net income = Income Before Taxes and Extraordinary Items - Income Taxes + Extraordinary items**
- **ROE Framework**
  - **ROE = Net Income / Total Equity Capital**
  - **ROE = (Net Income / Total Assets) * (Total Assets / Total Equity Capital)**
  - **ROE = ROA * EM** [ROA = Return on assets, EM = Equity Multiplier]
  - **ROA = (Net Income / Total Operating Income) * (Total operating income / Total Assets)**
  - **ROA = PM * AU** [PM = Profit margin, AU = Asset Utilization]
  - **AU = (Total Operating Income / Total Assets) = Interest Income Ratio + Non-int. income ratio**
- **NIM - Net Interest Margin**
  - **NIM = (Net Interest Income / Earning Assets)**
  - **NIM = (Interest Income - Interest Expense) / (Investment Securities + Net Loans and Leases)**
- **Spread:** Avg yield on earning assets - Avg cost of interest being liability
  - **Spread = (Interest Income / Earning Assets) - (Interest expense / Interest-bearing liabilities)**
- **Overhead Efficiency:** Ability to Generate Non-Interest Income To cover non-interest expenses
  - **OE = Non-interest income / Non-interest expense**
- **Large Banks**
  - Generally, you operate with lower amounts of equity capital than small banks.
  - Generally use more purchase funds and fewer core deposits.
  - Tend to put more into salaries, premises, and other expenses
  - Tend to diversify operations and services and also generate more non-interest income.

Risks Incurred By FIs

- **Credit Risk:** Promise Cash Flows from loans & securities held by FIs may not be paid in full.
  - FIs can diversify credit risk exposures by exploiting the law of large numbers.
  - Diversification reduces firm-specific credit risk. Does not reduce systemic credit risk.
- **Liquidity risk:** Sudden, unexpected inc. in liability withdrawals → asset liquidation at low prices
  - Asset side: Loan requests and exercise by borrowers of the loan commitments
  - Liability side: Several customers withdraw money from their depository accounts.
- **Interest rate risk:** Maturities of assets & liabilities are mismatched, and interest rates are volatile.
  - Asset transformation → FI buying primary securities & issuing secondary securities to fund
  - **Refinancing risk** → Risk that the cost of rolling over or re-borrowing funds will rise above the returns being earned on asset investments
  - **Reinvestment Risk** → Risk that returns on funds to be reinvested will fall below cost of funds.
  - **Price Risk** → Risk that the price of the security will change when interest rates change.
- **Market Risk:** Risk incurred in trading assets and liabilities due to changes in interest rates.
  - Closely related to interest rate and FX risk, adds another dimension of trading activity risk
  - Commercial and investment banks have seen a decline in income from traditional businesses, which has been matched by increased reliance on income from trading.
- **Off-Balance Sheet Risk:** Risk incurred as a result of OBS activities.
- **Foreign exchange risk:** Risk that exchange rate changes can affect the value of FI's assets and liabilities denominated in foreign currencies.
  - Net Long Position in Foreign Currency → FI has more foreign assets than liabilities.
    - FI loses when foreign currency falls relative to home currency.
  - Net short position in foreign currency → FI has less foreign assets than liabilities.
    - FI loses when foreign currency appreciates relative to home currency.
- **Country/Sovereign Risk:** Risk that repayments from foreign borrowers may be interrupted because of govt. interference
- **Tech & Operational Risk:** Risk that technological investments do not produce anticipated cost savings. Operational risk is risk that existing technology may malfunction or break down.
- **Digital disruption and fintech risk:** Risk that fintech firms could disrupt business (Lost Customers and Revenue)
- **Insolvency risk:** FI may not have even enough capital to offset a certain decline in the value of assets relative to liabilities. Consequence of one or more of the previous risks.

- **Altman's Z-Score: Z = (1.2 X_1) + (1.4 X_2) + (3.3 X_3) + (0.6 X_4) + (1.0 X_5)**
  - X1=(Working capital / total assets), X2=(Retained Earnings / Total assets), X3=(Earnings before Interest and Taxes / Fatal Assets), X4=(Mkt. value of equity / Book Value of Total Liab.), X5=(Sales / Total Assets)
  - Z < 1.81 → High Risk, 1.81 < Z < 2.99 → Indeterminate Risk, Z > 2.99 → Low Risk (DEFAULT)
- When a firm raises funds either by issuing bonds or increasing bank loans, it holds a very valuable default or repayment option. KMV Corp → **Expected default frequency (EDF): Probability that the market value of the firm's assets will fall below the promised repayments or debt liability in one year.**
- **Return on loan w/ ROA: (1+k) = 1 + ((f + BR + m) / (1 - b(1 - RR))**
  - **k** → Promised gross return on a loan per dollar lent; **f** → Loan fees charged by FI; **BR** → Base Benchmark Interest Rate; **m** → Spread or markup added to base rate; **b** → Compensating balance requirement percentage, **RR** → Reserve requirements set by regulators
- **Return on Loan w/ RAROC: RAROC = (One-Year Income on a loan) / (Loan Risk or Value at Risk)**
  - Loan is approved only if RAROC is sufficiently high relative to the benchmark return on equity.
  - Lending officer balances the loans' expected income against the loans' expected risk.

Managing Liquidity Risk

- **Liability side reasons:** Liability holders seek to cash in their financial claims immediately. FIs must borrow additional funds or liquidate assets.
- **Asset-side reasons:** Borrowers draw on loan commitments. FIs must run down cash assets, sell off other liquid assets, or borrow additional funds.
- DI's know that normally only a small portion of demand deposits will be withdrawn on any given day. Withdrawals are offset by new deposits.
- **FIs manage drain and pulse in two ways:**
  - **Purchase Liquidity Management:** Utilize markets for purchase funds. Interbank Markets for Short-Term Loans
  - **Stored Liquidity Management:** FI liquidated some of its assets.
- DI unused loan commitments grew from 529.4% to 1,000% from 1994 to 2008. Fell back to 608% during financial crisis.
- Determining financial gap is one way to measure liquidity risk exposure.
  - **Financial Gap = Average Loans - Average Deposits**
  - **Financial Gap = Borrowed funds - Liquid Assets**
  - **Fin. Gap + Liquid Assets = Fin. Requirement (Borrowed Funds)**
- **3 Sources of Liquidity:** Sell liquid assets such as T-bills; Borrow funds in the money funds market; Use excess cash reserves above min. reserve reqs.
- Another way to measure liquidity exposure is to compare key ratios with other peers of similar size and the same location.
- Liquidity index: Measures potential losses a DI could suffer
  - **I = sum of (w_i)*(Pi / P*i)** for all assets i = 1 to N
  - wi → Percentage of Asset in Portfolio, Pi = Price of DR liquidates asset today, P*i = Price of DR liquidates asset at some point in the future.
- **Liquidity risk measures implemented by BIS:**
  - BIS → Bank for International Settlements
  - Liquidity Coverage Ratio = (Stock of high-quality liquid assets / Total net cash outflows over next 30 calendar days) >= 100%
  - Net Stable Funding Ratio = (Available amount of stable fund / Required amount of stable fund) > 100%
- Liquidity planning allows managers to make important borrowing priority decisions. Lower cost of funding & Minimize excess reserves.
  - **D**elineation of managerial details and responsibilities; **D**etailed list of fund providers most likely to withdraw funds; **Id**entification of size of potential deposit and fund withdrawals over various time horizons in the future; **In**ternal limits on subsidiaries' and branches' borrowings, as well as acceptable risk premiums to pay in each market (Fed, Repo, CD, etc.)
- **Bank run** is sudden & unexpected increase in deposit withdrawals from DI.
- Demand deposits are first-come, first-served contracts.
- **Bank panic** is a systemic run on the parts of the banking industry as a whole.
- **Deposit insurance** created in the wake of bank panics. Coverage increased from $2,500 (1934) to $100,000 (1980) to $250,000 (2008). Primary intention is to deter runs and panics. Secondary objective is to protect the less informed saver against a reduction in wealth if they are last in line if DI fails.
- Fed provides discount window lending facility. Serves as lender of last resort.
- Interest rate at which securities are discounted is called discount rate.
- Three types: Primary, Secondary, and Seasonal.
- Life insurance companies hold cash reserves to meet policy payments and cancellation (surrender) payments. P&C companies have greater need for liquidity because claims are short term and unpredictable.
- Most states sponsor state guarantee funds for insurance companies.
  - Run and administered by the private insurance companies themselves.
  - No permanent guarantee fund exists except in New York. Size of required contributions that surveying insurers make varies widely from state to state.
  - Delay usually occurs before cash surrender values are received from the Guarantee Fund.
- Investment funds sell shares as a liability to investors and face similar liquidity problems when the number of withdrawals rises abnormally high.
- Incentives for deposit-like runs for investment fund shareholders are less.

Managing OBS Risk w/ Derivatives

- Managers are increasingly turning to OBS instruments such as forwards, futures, options, and swaps to hedge FI risks.
- **Spot contract** → agreement to transact with immediate exchange of assets.
- **Forward contract** → Agreement to transact with future exchange of set amount of assets at set price.
- **Futures Contract** → Agreement to transact with a future exchange of a set amount of assets for a price that is settled daily.
- **Microhedging** → Derivative securities contract to hedge a specific asset/liab.
- **Macrohedging** → Hedging the entire duration gap of an FI.
- Key is to take a position in the futures market to offset loss on the bal. sheet.
- Options: exchange-traded, OTC, embedded in secs, OR caps, collars, floors
- **4 strategies:** Buying a call option, Writing a call option, Buying a put option, Writing a put option (Buy call → bond price to rise, Buy put → price to fall)
- **Buy Cap** → Buying a call option or succession of call options on int. rates
- **Buy Floor** → Buying a put option on interest rates.
- **Buy Collar** → Simultaneous position. Sell a cap and Buy a floor.
- **Contingent credit risk with derivatives** → Counterparty defaults and payment obligations
  - Futures are default risk-free. Forwards have some default risk.
  - Standardized options are default risk-free. Specialized options have some
- **Interest rate swap** is used to swap a fixed interest rate for a floating rate.
- **Currency swaps** are used to hedge against foreign exchange rate risk.
- Credit default swaps allow FIs to better hedge credit risk.
  - Buyer of CDS makes periodic payments to seller until the end of the life of the swap or until the credit event specified in the contract occurs.
  - Credit swaps are important because credit risk is still more likely to cause an FI to fail than interest rate or FX risk.
  - CDS allows FI to maintain long-term customer lending relationships without bearing the full credit risk exposure.
- **Total return swap** involves an obligation to pay interest at a specified fixed or floating rate for payments representing the total return on a specified amount.
- **Pure Credit Swap** is a swap in which an FI receives the par value of the loan on default in return for paying a periodic swap fee.
- **Credit risk concerns with swaps:**
  - **Payment flows are interest, not principal.** Default risk on interest rate swaps is therefore less than on a regular loan.
  - **Stand by letters of credit.** Should default occur, standby letter of credit party would provide swap payments in lieu of default party.
  - **Netting and Swaps.** Generally, each party calculates the net difference between the fixed and floating payment, and one party makes a single payment for the net difference to the other party.

Managing Interest Rate & Insolvency Risk

- Interest risk takes place because of the mismatching of asset and liability maturities.
- **Repricing gap model** and **duration gap model** are used to measure interest rate exposure.
- If Fed wants to slow down the economy, it will tighten monetary policies by taking actions that increase interest rates. To stimulate the economy, it allows interest rates to fall.
- **Repricing Model**
  - Simple model used by small FIs in the US. A book value accounting cash flow analysis.
  - **Reprising Gap = Rate-sensitive assets (RSAs) - Rate-sensitive Liabilities (RSLs)**
  - If RSA < RSL (more short-term liabilities) → Refinancing Risk
  - If RSA > RSL (more short-term assets) → Reinvestment Risk

$$\Delta NII_i = (GAP_i)\Delta R_i = (RSA_i - RSL_i)\Delta R_i$$

ΔNII_i Change in net interest income in the ith maturity bucket.

GAP_i Dollar size of the gap between the book value of rate-sensitive assets and rate-sensitive liabilities in maturity bucket i.

ΔR_i Change in the level of interest rates impacting assets and liabilities in the ith maturity bucket.

Cumulative effect on a bank's net interest income may be estimated using:

$$\Delta NII = \left(\sum_{i=1\text{day}}^{1\text{year}} RSA_i - \sum_{i=1\text{day}}^{1\text{year}} RSL_i\right)\Delta R_i$$
$$= (CGAP)\Delta R_i$$

CGAP effect is the relation between changes in interest rates and changes in net interest income.

- **Spread effect** is the effect that a change in the spread between rates on RSA and RSLs has on Net Interest Income. $\Delta NII = (RSA \times \Delta R_{RSA}) - (RSL \times \Delta R_{RSL})$
- **4 weaknesses of the repricing model:** 1) Ignore market value effects of interest rate changes; 2) Ignores cash flow patterns within maturity bucket; 3) Fails to deal with the problem of rate insensitive asset and liability cash flow runoffs & prepayments; 4) Ignore OBS cash flows
- **Duration gap model**
  - Duration gap is a measure of overall interest rate risk exposure for an FI.

$$D = -\frac{\%\Delta \text{ in the market value of a security}}{\Delta R/(1+R)} \Delta A = A \times -D_A \times \frac{\Delta R}{(1+R)} \Delta L = L \times -D_L \times \frac{\Delta R}{(1+R)} \Delta E = -(D_A - kD_L) \times A \times \frac{\Delta R}{(1+R)}$$

  - Duration model is difficult to apply in the real world. Duration matching (restructuring balance sheet) can be costly. Immunization is a dynamic problem. Immunization is the process of matching the duration of assets to the duration of liabilities.
- **Insolvency Risk Management:** Primary means of protection is FI's equity capital.
- Economic meaning of capital is net worth. Difference between market value of assets and liab.
- **Book value of capital** is the difference between book value of assets and book value of liab.
- Book value relies on historical costs and accounting rules and does not reflect changes in market interest rates. Market value fluctuates with changing economic conditions and int. rates.
- Book value comprises of par value of shares, surplus value of shares, and retained earnings.
- The higher the interest rate volatility, the greater is the difference between market & book values.
- More frequent examinations and regulatory standards reduce discrepancy. The more assets that are traded, the easier it is to assess the true market value of the asset portfolio.
- **Market-to-book ratio** shows the discrepancy between the market value and book value.
- Market value accounting is difficult to implement, especially for small commercial banks with large amounts of non-traded assets.
- Market value accounting introduces an unnecessary degree of variability into FI's reported earnings. FI's are less willing to accept longer-term asset exposures if these assets must be continuously marked to market.

Hedging OBS Activities w/ Loan Sales & Securitization

- Loan sales involve splitting up larger loans and loan portfolios. Loan securitization involves the grouping of smaller loans into larger pools.
- Loan sale occurs when a financial institution originates a loan and sells it with or without recourse to an outside buyer.
- **Recourse** is the ability of a loan buyer to sell the loan back to the originator should it go bad.
- Correspondent banking involves small financial institutions making large loans and selling or syndicating parts of the loans to large financial institutions. Very similar to insurance.
- The large financial institutions may often sell parts of their loans, called participations, to smaller financial institutions. Similar to reinsurance.
- Syndicated loan market can be segmented into three categories:
  - Market makers are generally the large commercial banks that commit capital to create liquidity and take out direct positions in the market.
  - Active traders are mainly investment banks, commercial banks, and culture funds.
  - Occasional participants may be either sellers or buyers of loans.
- Participation in a loan is the act of buying a share in a loan syndication with limited contractual control and rights of the borrower.
- An assignment is the purchase of a share in a loan syndication with some contractual control and rights over the borrower.
- Three segments of the US loan sales market:
  - **Traditional Short-Term Loans:** Secured by assets of the borrowing firm and issued for a short term (90 days or less). Sold in units of one million and up.
  - **Highly leveraged transaction (HLT) loans:** Secured by assets of the borrowing firm. Long maturity: often 3 to 6 years
  - **Less developed country (LDC) loans** are made to less developed countries.
- Major buyers include **investment** banks, **vulture** funds (Specialized investment funds established to invest in distressed loans, often with an agenda that does not include helping the distressed firm survive), **other** domestic banks, **foreign** banks, **insurance** companies, **closed-end** bank loan mutual funds, **nonfinancial** corporations.
- Major sellers include money center banks, **small** regional or community banks, **foreign** banks, **investment** banks, **U.S.** government and its agencies
- Three market segments of the secondary market for less developed country loans include
  - **Sovereign bonds:** Government-issued debt
  - **Performing loans:** Original outstanding sovereign loans on which sovereign countries are currently maintaining promise payments
  - **Non-performing loans:** Secondary market prices for the sovereign loans of countries where there are no interest or principal events currently being made.
- **Factors Encouraging Future Loan Sales Growth:** Fee income, Liquidity risk, Capital Costs, Reserve Requirement
- **Factors Deterring Future Loan Sales Growth:** Access to the commercial paper market, Customer Relationship Effects, Legal Concern
- **Pass-through securities** offer investors an interest in a pool which is created by financial institutions pooling mortgages and other loans they originate. Usually a result of government-sponsored programs.
- Three government agencies are directly involved: Ginnie Mae (GNMA), Fannie Mae (FNMA), Freddie Mac (FHLMC)
- Mortgage loan securitization reduces the regulatory tax burden, interest rate risk exposure, and liquidity risk exposure that FI's face. Introduces a new risk called prepayment risk to the pass-through security holder. Pre-pay means to pay back a loan before its maturity to the FI that originated the loan. (Similar to reinvesting risk)
- Most fixed-rate mortgages are fully amortized (will be paid off within time).
- **Collateralized mortgage obligation** is a mortgage-backed bond issued in multiple classes or tranches. If borrowers prepay, CMO holders are paid back according to their class, so the first class will get paid first, which maximizes their prepayment risk, and the last class will have the least prepayment risk.
- Mortgage-backed bonds are bonds that are backed by a collateralized pool of assets, mostly mortgages. Asset-backed bonds are backed by a pool of different assets.
- Mortgage's backing MBBs remain on the balance sheet. Cash flows on mortgages backing the bond are not directly connected to the interest and principal payment on the MBB.